

PROJECT: Serverless Image/Video Processing

Services: S3, Lambda, Step Functions, Rekognition, SNS

Goal: Automatically process images or videos uploaded to S3 (e.g., resize, label detection).

Skills: Event-driven architecture, serverless orchestration.

Definition:

Serverless Image/Video Processing is a cloud-based solution that automatically processes images or videos when they are uploaded to storage, without requiring any server management. It uses services like Amazon S3 for storage, AWS Lambda for processing, AWS Step Functions for workflow management, Amazon Rekognition for detecting objects, and Amazon SNS for sending alerts.

Scope of the Project:

When images are uploaded to Amazon S3, the process begins with ease, an automated flow is triggered, running smoothly without any keys. With AWS Lambda handling tasks, no servers to maintain, and AWS Step Functions guiding steps, ensuring no delays or strain.

Methodologies:

- The project uses an event-driven approach, where uploading an image to Amazon S3 automatically triggers the workflow.
- AWS Lambda executes code in response to the S3 event without managing servers.
- AWS Step Functions is used to coordinate multiple steps in a defined sequence.
- Amazon Rekognition is used to analyze images and detect labels or objects.
- Processed results are stored back in S3 (output bucket) for further use.
- Amazon SNS sends notifications to users after processing is completed.
- Each component works independently and communicates through events, making the system loosely coupled.
- The system runs only when triggered, ensuring cost efficiency and scalability.

Services Used in this Project:

1. Amazon S3 (Simple Storage Service)

- Used to store input images/videos and output results
- Acts as the starting point (trigger) of the workflow

2. AWS Lambda

- Used to run code automatically when an event occurs
- Processes images and interacts with other AWS services
- No server management required

3. AWS Step Functions

- Manages and controls the workflow
- Executes multiple Lambda functions in sequence
- Ensures smooth and reliable execution

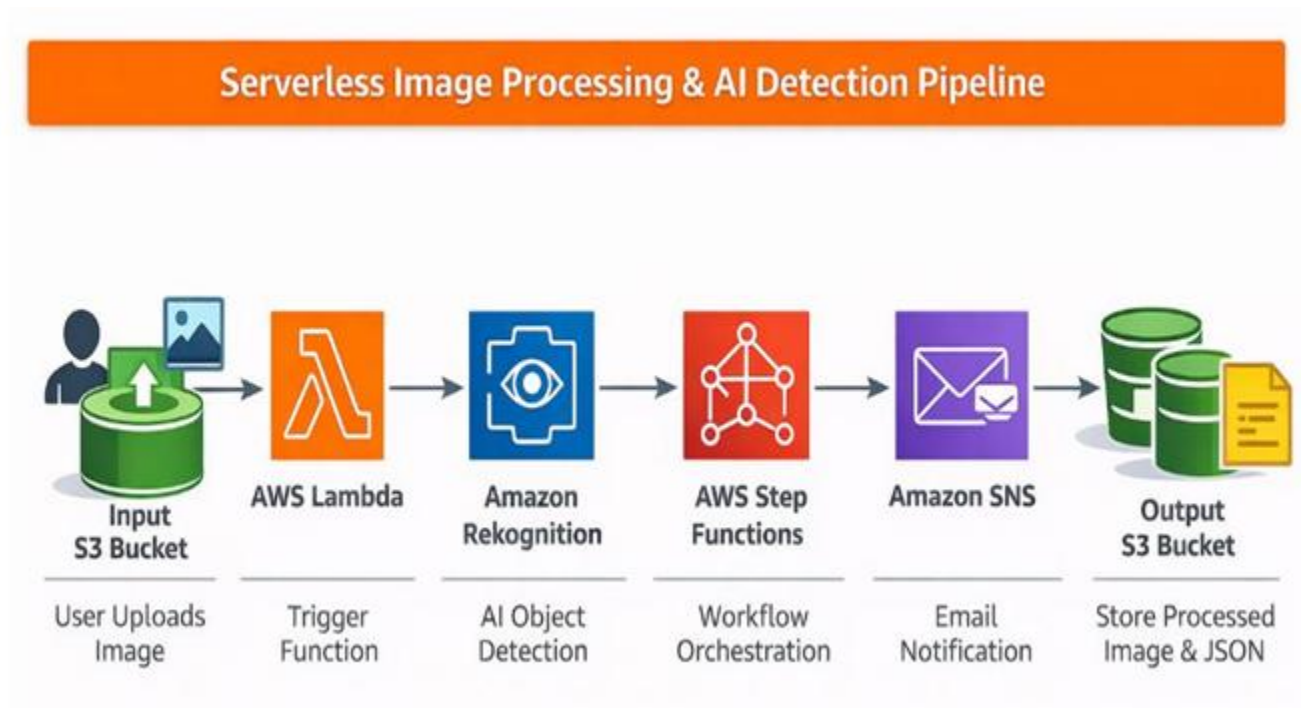
4. Amazon Rekognition

- Used for image and video analysis
- Detects objects, labels, and features in images
- Provides AI-based insights

5. Amazon SNS (Simple Notification Service)

- Sends notifications (email/SMS) to users
- Alerts users after image processing is completed

Architecture Diagram:



Implementation and Execution of Serverless Image/Video Processing

STEP 1: Created two S3 buckets

➤ Input Bucket: To upload images

Amazon S3

Buckets

General purpose buckets

Directory buckets

Table buckets

Vector buckets

Files

File systems

Access management and security

Access Points

Access Points for FSx

image-peocessing-input-shree-2026

ObjectsMetadataPropertiesPermissionsMetricsManagementFile systems - newAccess Points

Objects (1)

Copy S3 URICopy URLDownloadOpen L2DeleteActionsCreate folderUpload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Name	Type	Last modified	Size	Storage class
logo.jpg	jpg	April 9, 2026, 15:22:37 (UTC+05:30)	5.2 KB	Standard

➤ Output Bucket: To store processed images and results.

Amazon S3

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Access Points for FSx

Access Grants

IAM Access Analyzer

Storage management and

Buckets

General purpose bucketsAll AWS RegionsDirectory buckets

General purpose buckets (1/2)

Copy ARNEmptyDeleteCreate bucket

Buckets are containers for data stored in S3.

Find buckets by name

Name	AWS Region	Creation date
image-peocessing-input-shree-2026	Asia Pacific (Mumbai) ap-south-1	April 9, 2026, 13:44:44 (UTC+05:30)
image-peocessing-output-shree-2026	Asia Pacific (Mumbai) ap-south-1	April 9, 2026, 13:45:14 (UTC+05:30)

Account snapshot

Updated daily

View dashboard

Storage Lens provides visibility into storage usage and activity trends.

External access summary

Updated daily

External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

STEP 2: Created an IAM role and attached required policies.

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

User groups

Users

Roles

Policies

Identity providers

Account settings

Root access management

Temporary delegation requests

Access reports

Access Analyzer

Resource analysis

Unused access

Analyzer settings

Credential report

Organization activity

Service control policies

Resource control policies

ServerlessImageProcessingRole

Allows Lambda functions to call AWS services on your behalf.

Summary

Creation date: April 09, 2026, 13:47 (UTC+05:30)

Last activity: 23 minutes ago

ARN: arn:aws:iam:074139044318:role/ServerlessImageProcessingRole

Maximum session duration: 1 hour

PermissionsTrust relationshipsTagsLast AccessedRevoke sessions

Permissions policies (4)

Simulate L2RemoveAdd permissions

You can attach up to 10 managed policies.

Filter by Type: All types

Policy name	Type	Attached entities
AmazonRekognitionFullAccess	AWS managed	1
AmazonS3FullAccess	AWS managed	3
AmazonSNSFullAccess	AWS managed	1
AWSStepFunctionsFullAccess	AWS managed	1

STEP 3: Created a Lambda function and added Python code to read images from the S3 input bucket and process them.

The screenshot shows the AWS Lambda console and an IDE. The console displays a list of functions: processimage, saveresult, and sendNotification. The IDE shows the code for the processimage function, which uses boto3 to interact with the Rekognition service to detect labels in an image from an S3 bucket.

Functions (4) Last fetched 4/9/2026, 3:47:22 PM

Function name	Description	Package type	Runtime	Type	Last modified
processimage	-	Zip	Python 3.14	Standard	2 hours ago
saveresult	-	Zip	Python 3.14	Standard	1 hour ago
sendNotification	-	Zip	Python 3.14	Standard	2 hours ago

```
1 import boto3
2
3 rekognition = boto3.client('rekognition')
4
5 def lambda_handler(event, context):
6     bucket = event['bucket']
7     key = event['key']
8
9     response = rekognition.detect_labels(
10         Image={
11             'S3Object': {
12                 'Bucket': bucket,
13                 'Name': key
14             },
15         },
16         MaxLabels=5
17     )
18
19     labels = [label['Name'] for label in response['Labels']]
20
21     return {
22         "bucket": bucket,
23         "key": key,
24         "labels": labels
25     }
```

STEP 4: Created an SNS topic and subscription to send email notifications with the image analysis results.

➤ Created Topic

The screenshot shows the Amazon SNS console. The topic 'ImageProcessingAlerts' is selected, and its details are displayed. A subscription is listed with the endpoint 'srinivassindhanur7@gmail.com' and status 'Confirmed'.

Amazon SNS > Topics > ImageProcessingAlerts

ImageProcessingAlerts Edit Delete Publish message

Details

Name	ARN	Display name	Type
ImageProcessingAlerts	arn:aws:sns:ap-south-1:074139044318:ImageProcessingAlerts	-	Standard

Topic owner: 074139044318

Subscriptions (1) Edit Delete Request confirmation Confirm subscription Create subscription

ID	Endpoint	Status	Protocol
ab9b1610-84a6-4a14-90e2-5fa1...	srinivassindhanur7@gmail.com	Confirmed	EMAIL

➤ Created Subscription

Amazon SNS > Subscriptions

Subscriptions (1)

Search

ID	Endpoint	Status	Protocol	Topic
ab9b1610-84a6-4a14-90e...	srinivassindhanur7@gmail....	Confirmed	EMAIL	ImageProcessingAlerts

Buttons: Edit, Delete, Request confirmation, Confirm subscription, Create subscription

STEP 5: Created a Step Functions state machine to orchestrate the workflow and invoke the Lambda function.

Step Functions > State machines > mystatemachine > Execution: edef9bb3-b8b6-4ae9-9f03-bdce2dc63330

Execution: **edef9bb3-b8b6-4ae9-9f03-bdce2dc63330**

Buttons: Edit state machine, New execution, Actions

Details | Execution input and output | Definition

Execution status: **Succeeded**

Execution type: Standard

Execution ARN: [arn:aws:states:ap-south-1:074139044318:execution:mystatemachine:edef9bb3-b8b6-4ae9-9f03-bdce2dc63330](#)

IAM role ARN: [arn:aws:iam::074139044318:role/service-role/StepFunctions-mystatemachine-role-4u6xb3uj2](#)

State transitions: [Learn more](#) 5

Start time: Apr 9, 2026, 15:22:38.876 (UTC+05:30)

End time: Apr 9, 2026, 15:22:42.382 (UTC+05:30)

Duration: 00:00:03.506

Alias: -

Version: -

Buttons: Graph view, Table view

Step Functions > Graph view

Graph view

Start

AWS Lambda: Invoke ProcessImage

AWS Lambda: Invoke SaveResult

AWS Lambda: Invoke SendNotification

End

ProcessImage

Logs | Lambda | Log group

Input/Output | Variables | Details | Definition | Events

Advanced view

State input

```
1 {
2   "bucket": "image-peocessing-input-shree-2026",
3   "key": "logo.jpg"
4 }
```

State output

```
1 {
2   "bucket": "image-peocessing-input-shree-2026",
3   "key": "logo.jpg",
4   "labels": [
5     "Logo",
6     "Text",
7     "Electronics",
8     "Computer"
9   ]
10 }
```

Legend: In progress, Failed, Caught error, Canceled, Succeeded

STEP 6: Tested the workflow by uploading an image to the S3 input bucket and verifying the output image and notification.

Amazon S3

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Access Points

Access Points for FSx

image-processing-output-shree-2026

Objects

Metadata

Properties

Permissions

Metrics

Management

File systems - new

Access Points

Objects (1)

Copy S3 URI

Copy URL

Download

Open

Delete

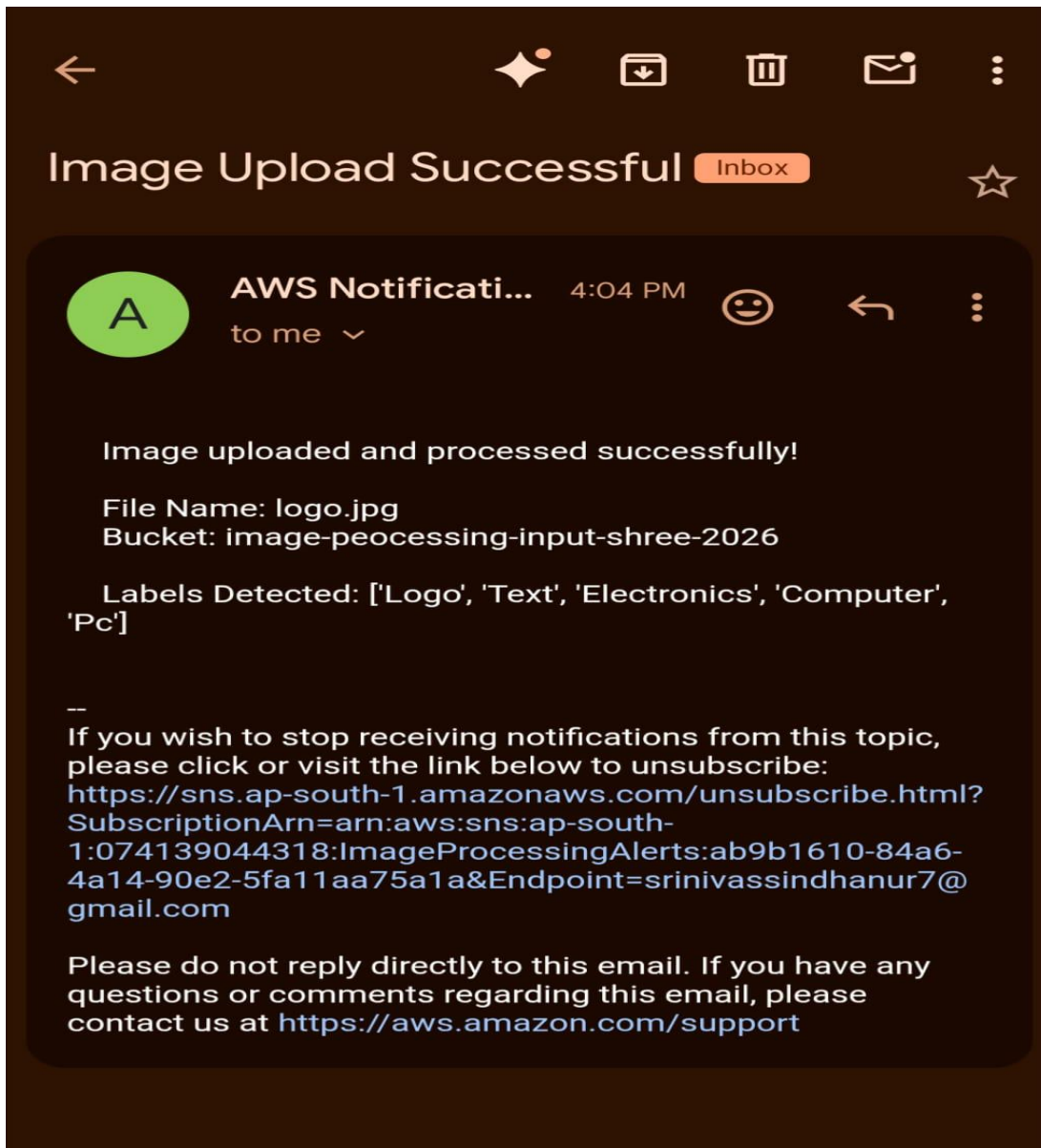
Actions

Create folder

Upload

Find objects by prefix

	Name	Type	Last modified	Size	Storage class
☐	logo.jpg.json	json	April 9, 2026, 15:22:42 (UTC+05:30)	127.0 B	Standard



Troubleshoot:

- No troubleshoot

Conclusion:

This project demonstrates how AWS serverless services can be used to build an automated image processing system. By integrating S3, Lambda, Rekognition, Step Functions, and SNS, the system analyzes images and sends notifications without managing servers. The workflow is event-driven, scalable, and cost-efficient because it runs only when an image is uploaded. This project highlights the power of cloud automation and AI services in building modern, efficient applications.

